

ROBOTICS

Engineering the Structural Logic of Autonomous Robotics & Automation

A project-first certification designed to master ROS 2 (Robot Operating System), SLAM algorithms, and the structural engineering of multi-axis robotic systems.



THE EDLANX METHODOLOGY

OUR ROBOTICS ENGINEERING PHILOSOPHY

At Edlanx, we don't just teach "building toys"; we cultivate a robotics-architect mindset. Our program is a deep dive into the synergy of sensors, actuators, and software. We believe in Precision Autonomy — moving beyond basic remote-controlled systems to build robust, self-navigating machines that ensure safety and efficiency in industrial, medical, and delivery applications.

THE FOUR PILLARS OF MASTERY

-  **Kinematic Logic** Master the core mathematics of forward and inverse kinematics for precise motion.
-  **Perception Governance** Learn the structural logic of LiDAR, Computer Vision, and sensor fusion.
-  **Control Architecture** Understand the engineering of PID loops, trajectory planning, and motor synchronization.
-  **Autonomous Integration** Bring your machines to life by mastering the structural link between ROS 2 nodes and physical hardware.

Table of Contents

Program Overview

The Edlanx Methodology

Our Development Philosophy

The Edlanx Advantage

01

The System Blueprint: Robotics Architecture

1.1 Component Logic

1.2 Communication Blueprinting

02

Kinematic Architecture: Motion Mathematics

2.1 Coordinate Logic

2.2 Forward & Inverse Kinematics

2.3 Degrees of Freedom (DoF) Governance

2.4 Trajectory Blueprinting

03

Actuation Logic: Motors & Drives

3.1 Servo & Stepper Architecture

3.2 Torque & Speed Governance

3.3 Feedback Architecture

3.4 H-Bridge & PWM Logic

3.5 Harmonic Drive Engineering

3.6 Power Distribution Governance

04

The Operating System: ROS 2 Architecture

4.1 Node & Topic Logic

4.2 Service & Action Governance

4.3 Workspace Engineering

05

Perception Architecture: Sensors & Vision

5.1 Sensor Fusion Logic

5.2 LiDAR & Ranging Governance

06

Navigation Logic: Path Planning & SLAM

6.1 SLAM Architecture

6.2 Probabilistic Logic

6.3 Global Path Planning Governance

6.4 Local Planner Engineering

6.5 Costmap Architecture

6.6 Behavior Tree Governance

6.7 Odometry Integration Logic

07

Control Architecture: PID & State Space

7.1 PID Governance

7.2 State-Space Logic

7.3 Feedback Loop Architecture

08

Industrial Robotics: PLC & Cobot Integration

8.1 Cobot Governance

8.2 PLC Logic Integration

8.3 End-Effector Architecture

09

Simulation Logic: Gazebo & URDF Modeling

9.1 URDF Blueprinting

9.2 Gazebo Environment Logic

9.3 "Hardware-in-the-Loop" Simulation Architecture

10

Embedded Robotics: Real-Time Hardware Interfacing

10.1 Real-Time Kernel Logic

10.2 Communication Bus Architecture

10.3 Micro-Architectural Interfacing

11

Swarm Logic: Multi-Robot Coordination

11.1 Decentralized Governance

11.2 Multi-Agent Communication Logic

11.3 Collaborative Task Architecture

12

Future Frontiers: Humanoid Logic & AI Robotics

12.1 Bipedal Stability Architecture

12.2 Foundation Model Integration



WHY CHOOSE EDLANX



01 Industry-Ready Curriculum

We move beyond building basic hobbyist kits to bridge the gap between "toy robotics" and high-level industrial systems. Our Robotics & Automation syllabus is a deep immersion into ROS 2 (Robot Operating System), SLAM algorithms, and the complex kinematic mathematics required by global manufacturing, logistics, and R&D giants.

02 Real-Time Corporate Exposure

Step into a high-performance professional environment at our Bangalore Operational Center. At Edlanx, you won't just move a servo; you will be treated as a Robotics Systems Engineer, working within real organizational frameworks to architect autonomous navigation stacks and manage multi-axis robotic coordination.

03 Mentorship & Continuous Assessment

Your growth is guided by senior Robotics Architects and Automation Leads through structured mentoring and rigorous performance evaluations. We focus on developing your ability to solve complex coordinate transformations, sensor-fusion hurdles, and real-time control conflicts with the precision required for production-ready machines.

04 Skill Mastery & Performance Incentives

Master industry-standard tools and platforms like ROS 2, Gazebo, RViz, and MATLAB while learning essential professional competencies. We believe in experiential learning and reward innovative autonomous path-planning or hardware-integration breakthroughs with performance-based stipends, ensuring you remain driven toward technical mastery.

05 Global Career Readiness

We enhance your employability by focusing on the complete Robotic Development Lifecycle, from initial URDF modeling to final physical deployment and diagnostic validation. Our career guidance includes technical resume engineering and mock "System Design Reviews" designed to help you secure elite roles in global R&D centers.

06 Collaborative Engineering Community

Join a supportive culture built on transparency, integrity, and disciplined professional ethics. You will learn to lead a robotics "sprint," collaborate across mechanical, electronic, and software domains, and grow your network within an elite community of engineers, developers, and mentors.

07 Verified Professional Certifications

At Edlanx, we don't just issue certificates for attendance; we award them based on verified learning, real-world performance, and practical outcomes. Our credentials are proof of your industry-ready skills and are recognized by institutions and recruiters worldwide.

We provide four distinct levels of certification:

- Internship Completion Certificate: Validates your exposure to professional work culture, verified internship hours, and task-based mentorship.
 - Project Completion Certificate: Proof of your hands-on implementation and problem-solving skills in real-world industry projects.
 - Training Completion Certificate: Issued upon mastering our industry-aligned curriculum and passing skill-based assessments.
 - Certificate of Excellence: A prestigious award reserved for top-performing candidates who demonstrate an advanced understanding.
-

Why This Journey Matters

- ▶ **Portfolio-Ready** Every autonomous navigation stack and kinematic control algorithm you architect at Edlanx serves as a live, audit-ready demonstration of your ability to manage the physical and digital complexity of modern robotics.
- ▶ **Global Network** Gain exclusive access to the Edlanx alumni community and industry mentors leading R&D teams at global robotics labs, industrial giants, and pioneering deep-tech startups.
- ▶ **Career Excellence** We move beyond basic assembly to provide the technical resume engineering and system-integration logic needed to lead the future of "Intelligent Machines" and the global transition to fully autonomous industry.

This course is a living ecosystem. We constantly update our curriculum to reflect the latest shifts in ROS 2 Humble/Iron standards, Generative AI for Robotics (VLA models), and Human-Robot Collaboration protocols. Your success is our shared mission.

DIRECT MENTORSHIP



Engineering a reliable autonomous system requires absolute mathematical and spatial precision. Whether you are debugging a SLAM drift issue, optimizing a 6-DoF inverse kinematic solver, or architecting a multi-robot swarm communication protocol, our senior Robotics Architects are available to provide the direct, technical feedback needed to ensure your systems are production-ready.

Address **Lotus Square, 9th cross Rd, 7th Sector, HSR Layout, Bengaluru**

Phone **+91 93608 40496**

Email **www.edlanx.com
support@edlanx.com**